

Sales Data Analysis Report

Introduction

This project analyzes a sales dataset using Python (Pandas, Matplotlib, and Seaborn) and Power BI. The objective was to identify sales trends, top-performing products and cities and other key business insights.

Data Preparation

The dataset was cleaned by removing unnecessary columns and converting the Order Date column into a proper date format. Additional columns such as Year, Month Name, Day and Weekday were created to support the analysis.

Key Findings

- San Francisco recorded the highest sales among all cities.
- MacBook Pro Laptop generated the highest sales revenue among all products.
- December was the highest-performing month in terms of sales.
- Tuesday recorded the highest sales among all weekdays.
- Sales were significantly higher in 2019 compared to 2020; however the 2020 data appears incomplete and should be interpreted with caution.
- Analysis of sales patterns revealed periods of increased customer purchasing activity, providing useful insights for future business planning.

Report On Sales Trend

The sales trend analysis revealed that 2019 recorded total sales of \$34,483,365.68, making it the highest-performing year in the dataset. In comparison, 2020 generated sales of only \$8,670.29.

The significant difference between the two years is primarily due to the number of records available. The dataset contains 185,916 transactions for 2019 but only 34 transactions for 2020.

This indicates that the 2020 data is incomplete and should not be used to draw conclusions about business performance.

Based on the available data, the company experienced strong sales performance throughout 2019. The apparent decline in 2020 does not necessarily represent a reduction in demand or revenue but rather reflects limited data availability. Therefore, 2019 provides the most reliable basis for further sales analysis and business insights.

Conclusion

The analysis revealed important sales patterns across products, cities, months and years. The findings can help support data-driven decision-making in areas such as inventory planning, marketing campaigns and sales strategy. Power BI dashboards and Python visualizations were used to effectively communicate these insights.